*Universidad Politecnica de Madrid*

# Embedded System Design with Raspberry Pi

**Student:** Blanca Loma Mecha

**Course:** Embedded System Design with Raspberry Pi

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**Instructor:** Mariano Ruiz

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## Contents

## 1. Project 1: Getting started with the Raspberry-Pi and Raspbian

### 1.1 Objectives

- Learn how to use the RPI
- Develop a C file using I2C to communicate with the accelerometer sensor
- Learn how to use AI tools to resolve problems

### 1.2 Week 1

#### Task List

| ID | Description | Owner | Status |
|---|---|---|---|
| W1-T1 | Install Ubuntu 22.04 on personal laptop | Blanca Loma | Completed |
| W1-T2 | Create git account and repository, and install git in Ubuntu | Blanca Loma | In Progress |
| W1-T3 | Answer the questions from pages 33-34 of RPI Introduction | Blanca Loma | In Progress |
| W1-T4 | Install Raspbian on the SD Card | Blanca Loma | Completed |
| W1-T5 | Learn how to connect the sensor | Blanca Loma | In progress |

#### Work Log

**2026-02-22 — Work session**

- I successfully installed Raspbian following the instructions in the provided page.
- I created a git account.
- I installed correctly Ubuntu and tried to complete all the git tasks
- I started to answer some of the questions
- Sofi and I met to find information on how to connect the sensor

**2026-02-22 — Debugging**

• I tried to install Ubuntu, but it stopped downloading. I will try on another computer.

• I had a problem trying to copy the repository. There was an error indicating that authentication is not supported for Git operations.

### 1.3 Week 2

## Task List

| ID | Description | Owner | Status |
|---|---|---|---|
| W2-T1 | Create SSH keys in every laptop I use | Blanca Loma | Completed |
| W2-T2 | Connect the sensor | Blanca Loma | Completed |
| W1-T3 | Answer the questions from pages 33-34 of RPI Introduction | Blanca Loma | In Progress |
| W1-T4 | Create the code for the sensor | Blanca Loma | In Progress |

## Work Log

**2026-02-25 — Work session**

• I created the SSH keys in all the computers I need to use for the rest of the course

• My teammate and I figured out how to connect the sensor

• My teammate and I tried the Raspbian and connected everything

• I tried to start the sensor code

**2026-02-25 — Debugging**

• We had some troubles because of the lack of internet in the class. We couldn't copy the repository on the SD operative system, so I had to do it at home.

## 1.4 Week 3

### Task List

| ID | Description | Owner | Status |
|---|---|---|---|
| W3-T1 | Answer the questions from pages 33-34 of RPI Introduction | Blanca Loma | Completed |
| W3-T2 | Create the code for the sensor | Blanca Loma | Completed |

### Work Log

**2026-03-10 — Work session**

• I finished answering the questions

• We found how to prove the sensor was recognized, funding the address we needed for developing the code

• We finished the code for the sensor

**2026-03-10 — Debugging**

• We had some problems because we had to wake up the 107 register, but we managed to solve it.

• We also had problems with the variables because they were in two’s complement format.

## 2. Project 2: Using Buildroot for building Embedded Linux System on RPI-4

• Learn another way to use the RPI, by using Buildroot
• Learn how to change Buildroot configuration when needed
• Develop a C file for the colour sensor and the accelerometer

### 2.2 Week 1

**Task List**

| ID | Description | Owner | Status |
|---|---|---|---|
| W4-T1 | Create a new git repository for the Project II progress | Blanca Loma | Completed |
| W4-T2 | Reuse the C code developed in Project I for the accelerometer to use the colour sensor | Blanca Loma | In Progress |
| W4-T3 | Build Linux using buildroot | Blanca Loma | In Progress |

**Work Log**

**2026-03-17 — Work session**

• We finished the 3.6 point of the manual for building Linux using buildroot.

### 2.3 Week 2

**Task List**

| ID | Description | Owner | Status |
|---|---|---|---|
| W5-T1 | Build Linux using buildroot | Blanca Loma | In Progress |
| W5-T2 | Reuse the C code developed in Project I for the accelerometer to use the colour sensor | Blanca Loma | In Progress |

**Work Log**

**2026-03-24 — Work session**

• We finished the 3.9 point of the manual for building Linux using buildroot

**2026-03-24 — Debugging**

• We didn't understand the method to add WIFI support at first, but we have understanded it now

## 2.4 Week 3

### Task List

| ID | Description | Owner | Status |
|---|---|---|---|
| W6-T1 | Build Linux using buildroot | Blanca Loma | Completed |
| W6-T2 | Reuse the C code developed in Project I for the accelerometer to use the colour sensor | Blanca Loma | Completed |
| W6-T3 | Implement an application to measure from the colour sensor | Blanca Loma | Completed |
| W6-T4 | Combine both codes | Blanca Loma | Completed |
| W6-T5 | Create the presentation | Blanca Loma | Completed |

### Work Log

**2026-04-01 — Work session**

• First we finished building our Linux using Buildroot. We had to repeat the process of creating the image 3 times, because the basic configuration that we followed in the 3.0 to 3.6 steps of the PDF needed also to configure the Wi-fi support and the I2C.

• When we finished the configuration, we started to boot the RPI, introducing the SD card and connecting the FTDI terminal and the power supply. Then we started the putty program with the configuration we needed  and after changing the PermitRootLogin parameters of the sshd_config file, to “yes”, we were able to enter as root with SSH.

• We configured the cross-compilation by following the 6.3 step of the PDF. We created a Makefile and a .vscode folder with four .json files. Also we reused the .c code from Project I. Finally by running the VSCode we were able to use our compiled file in the terminal.

• We added the code for the accelerometer. First we found in the TCS3472 datasheet that we had to configure some parameters, and we did it like in the Project I. And finally we started the data loop, reading from the 0x14 addr (it is important that we needed to add to this address the command bit 0x80).

• Finally we combined both codes from both sensors, and started to measure at the same time.

• For the presentation, we had every step written in a pdf we created at the beginning, so it was very easy to resume everything.

**2026-04-01 — Debugging**

• We had some problems because we configured everything in the class computer and we weren’t able to copy the rpi folder, so this made us go to the lab a lot of times, because we couldn’t change the image file or compile the code without it.

## 3. Project 3: IOT Application with Raspberry Pi

### 3.1 Objectives

• Learn how to implement a client-server project for the RPI
• Learn how to implement the code for the server and the code for the client
• Develop the integration with ThingsBoard

### 3.2 Week 1

**Task List**

| ID | Description | Owner | Status |
|---|---|---|---|
| W1-T1 | Develop the client-server code | Blanca Loma | In Progress |
| W1-T2 | Install Eclipse | Blanca Loma | Completed |

### 3.3 Week 2

**Task List**

| ID | Description | Owner | Status |
|---|---|---|---|
| W2-T1 | Develop the client-server code | Blanca Loma | Completed |

### 3.4 Week 3

**Task List**

| ID | Description | Owner | Status |
|---|---|---|---|
| W3-T1 | Compile the code in eclipse to correct the errors | Blanca Loma | Completed |
| W3-T2 | Using putty to debug and execute the application | Blanca Loma | Completed |

## Work Log

### 2026-05-06 — Debugging

• We had some problems trying to debug the code, because the RPI couldn't detect the i2c device at one point, and we found out that the config.txt file didn't have the parameters to allow the i2c. After changing it and rebooting the RPI, we were able to debug our code correctly.

## 3.5 Week 4

## Task List

| ID | Description | Owner | Status |
|---|---|---|---|
| W4-T1 | Prepare the presentation | Blanca Loma | Completed |
| W4-T2 | Finish the mandatory part | Blanca Loma | Completed |
| W4-T3 | Optional part | Blanca Loma | Completed |

## Work Log

### 2026-05-13 — Work session

• We finished the mandatory part, we only needed to make the client start automatically, by implementing a file named with an specific initial part of the name (S99) so it can be initiated when the RPI starts.

• We started taking captures for the presentation, and sending to git the final files.

• I started to install ThingsBoard and created the temperature device from the tutorial

### 2026-05-19 — Work session

• We installed the mosquitto client on the RPI, and proved that we could send telemetries from the RPI.

• We changed the IOTCLient.c file to send the telemetries with the samples we take every second, and prove that telemetries changed every second in ThingsBoard.

• We created a Dashboard with two time series charts, one of accelerometer and the other for the colour sensor, and a radar that is useful to see a 3D figure based on the accelerometer movement.

- We finished the presentation, adding the ThingsBoard part, and including the last captures.